

L^AT_EX Table Practice Pack

10 Graded Exercises Modelled on the Lab-Exam Papers

Reproduce Part I from scratch — check Part II afterwards

July 21, 2026

Rules of engagement. Print Part I only. Start from a blank file with your own preamble. Time each attempt. A table “passes” only if it *compiles* and matches the target structurally — merged cells in the right places, rules in the right places, maths rendered as maths.

Contents

PART I — Target Tables	2
Exercises 1–10	2
PART II — Solutions and Marking Notes	8
Solutions 1–10	8
Drill Routine	14
The Five Errors That Cost the Most Marks	15

PART I — Target Tables

Reproduce each of the following

Exercise 1 — Warm-up

Difficulty: ★ **Target time:** 3 minutes
Tests: column spec, borders, inline maths in cells

Table 1: Sorting-algorithm complexity.

Algorithm	Best	Average	Worst
Bubble sort	$\mathcal{O}(n)$	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$
Merge sort	$\mathcal{O}(n \log n)$	$\mathcal{O}(n \log n)$	$\mathcal{O}(n \log n)$
Quick sort	$\mathcal{O}(n \log n)$	$\mathcal{O}(n \log n)$	$\mathcal{O}(n^2)$
Binary search	$\mathcal{O}(1)$	$\mathcal{O}(\log n)$	$\mathcal{O}(\log n)$

Exercise 2 — Two-tier header

Difficulty: ★★ **Target time:** 5 minutes
Tests: multicolumn, multirow, cline — the header pattern from the Laplace paper

Table 2: Numerical integration methods.

Method	Error		Order
	Local	Global	
Trapezoidal	$\mathcal{O}(h^3)$	$\mathcal{O}(h^2)$	2
Simpson	$\mathcal{O}(h^5)$	$\mathcal{O}(h^4)$	4
Euler	$\mathcal{O}(h^2)$	$\mathcal{O}(h)$	1

Exercise 3 — Multirow families with stacked fractions

Difficulty: ★★★ **Target time:** 8 minutes
Tests: multirow spanning 3 rows, dfrac in cells, arraystretch, partial cline

Table 3: Standard derivatives and integrals by family.

Family	Operation		Domain
	Function	Derivative	
Power	x^n	nx^{n-1}	$n \in \mathbb{R}$
	$\frac{1}{x}$	$-\frac{1}{x^2}$	$x \neq 0$
	$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	$x > 0$
Trigonometric	$\sin x$	$\cos x$	$x \in \mathbb{R}$
	$\tan x$	$\sec^2 x$	$x \neq \frac{\pi}{2}$
Exponential	e^{ax}	ae^{ax}	$a \in \mathbb{R}$
	$\ln x$	$\frac{1}{x}$	$x > 0$
Chain rule $f(g(x))$		$f'(g(x)) g'(x)$	differentiable

Exercise 4 — Coloured table with scientific notation

Difficulty: *** **Target time:** 8 minutes

Tests: rowcolor, cellcolor, fraction inside a header cell, $\times 10^{-n}$ notation

Table 4: Semiconductor material parameters.

Material	Electrical		Derived	
	E_g (eV)	μ_n	$\sigma = \frac{1}{\rho}$	$\beta = \sqrt{\mu_n}$
Silicon	1.12	1.35×10^3	4.35×10^{-4}	36.7
Germanium	0.67	3.90×10^3	2.17×10^{-2}	62.4
GaAs	1.42	8.50×10^3	1.00×10^{-8}	92.2
InP	1.34	5.40×10^3	1.25×10^{-7}	73.5
Mean gap			1.14 eV	

Exercise 5 — Spanning in both directions

Difficulty: *** **Target time:** 7 minutes

Tests: multirow and multicolumn interacting, merged footer row

Table 5: Course assessment breakdown.

Component	Item	Weight	Marks
Continuous	Quizzes	10%	10
	Assignments	15%	15
	Lab work	15%	15
Examination	Midterm	25%	25
	Final	35%	35
Total		100%	100

Exercise 6 — Professional rules with wrapped cell text

Difficulty: ** **Target time:** 6 minutes

Tests: booktabs, makecell line breaks inside a cell, no vertical rules

Table 6: Comparison of transform methods.

Transform	Kernel function	Typical application
Laplace	e^{-st}	transient circuits, control systems
Fourier	$e^{-j\omega t}$	spectra, signal filtering
Z	z^{-n}	discrete systems, digital filters
<i>All three convert differential relations into algebraic ones.</i>		

Exercise 7 — Fixed total width with a wrapping column

Difficulty: ★★ **Target time:** 5 minutes

Tests: tabularx, the X column, maths inside a paragraph column

Table 7: Boundary conditions and their meaning.

Name	Form	Physical interpretation
Dirichlet	$u = g$ on $\partial\Omega$	The value of the solution is prescribed on the boundary, for example a surface held at a fixed temperature.
Neumann	$\frac{\partial u}{\partial n} = g$	The flux across the boundary is prescribed; a zero value models a perfectly insulated wall.
Robin	$\alpha u + \beta \frac{\partial u}{\partial n} = g$	A weighted combination of the two above, used for convective heat exchange with a surrounding medium.

Exercise 8 — Heavy maths inside cells

Difficulty: ★★★ **Target time:** 10 minutes

Tests: matrices, cases, integrals and limits nested inside table cells

Table 8: Objects that must survive inside a cell.

Object	Expression	Note
Matrix	$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$	2×2 real
Determinant	$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$	identity
Piecewise	$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}$	continuous at 0
Integral	$\int_0^\infty e^{-st} dt = \frac{1}{s}$	$s > 0$
Limit	$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$	classical
Nested index	$\sum_{\substack{i=1 \\ i \neq j}}^n a_{ij}$	double subscript

Exercise 9 — Three-tier header with a diagonal cell

Difficulty: ★★★★★ **Target time:** 12 minutes
Tests: diagbox, a multirow spanning three header rows, repeated multicolumn tiers

Table 9: Network layer performance matrix.

Metric Layer	Training		Inference	
	(per epoch)		(per sample)	
	Time	Memory	Time	Memory
Convolution	12.4 s	1.8 GB	3.1 ms	0.4 GB
Pooling	0.8 s	0.2 GB	0.2 ms	0.1 GB
Dense	4.6 s	3.4 GB	1.0 ms	0.9 GB
Total	17.8 s	5.4 GB	4.3 ms	1.4 GB

Exercise 10 — Full exam simulation

Difficulty: ★★★★★ **Target time:** 15 minutes
Tests: everything at once — colour, both spans, nested maths, footnote marker, caption, label and a cross-reference from the body text

Table 10 summarises the four candidate designs.

Table 10: Design trade-off study for the RC filter stage.

Design	Components		Response	
	$R \text{ (k}\Omega\text{)}$	$C \text{ (}\mu\text{F}\text{)}$	$\tau = RC$	$f_c = \frac{1}{2\pi RC}$
Fast	1.0	0.10	0.10 ms	1.59 kHz
	2.2	0.10	0.22 ms	723 Hz
Slow	10.0	1.00	10.0 ms	15.9 Hz
	22.0	4.70	103 ms	1.54 Hz ¹
Selected design			Fast, $R = 1.0 \text{ k}\Omega$	

The selected design in Table 10 gives the widest bandwidth.

¹Measured with a 5 V step input at 25°C.

PART II — Solutions and Marking Notes

Do not read until you have attempted Part I

Solution 1

```
\begin{table}[H]
\centering
\caption{Sorting-algorithm complexity.}
\begin{tabular}{|l|c|c|c|}
\hline
\textbf{Algorithm} & \textbf{Best} & \textbf{Average} & \textbf{Worst} \\
\hline
Bubble sort &  $\mathcal{O}(n)$  &  $\mathcal{O}(n^2)$  &  $\mathcal{O}(n^2)$  \\
\hline
Merge sort &  $\mathcal{O}(n \log n)$  &  $\mathcal{O}(n \log n)$  &  $\mathcal{O}(n \log n)$  \\
\hline
Quick sort &  $\mathcal{O}(n \log n)$  &  $\mathcal{O}(n \log n)$  &  $\mathcal{O}(n^2)$  \\
\hline
Binary search &  $\mathcal{O}(1)$  &  $\mathcal{O}(\log n)$  &  $\mathcal{O}(\log n)$  \\
\hline
\end{tabular}
\end{table}
```

Marking notes. The column spec `{|l|c|c|c|}` has four entries and four `|` fences — count them before typing the body. `\log` must be a command, not the letters `log`, or it prints in italic. `\mathcal{O}` needs `amssymb/amsfonts`.

Solution 2

```
\begin{table}[H]
\centering
\caption{Numerical integration methods.}
\renewcommand{\arraystretch}{1.3}
\begin{tabular}{|l|c|c|}
\hline
\multirow{2}{*}{\textbf{Method}} & \multicolumn{2}{c}{\textbf{Error}} \\
\cline{2-3}
& \textbf{Local} & \textbf{Global} \\
\hline
Trapezoidal &  $\mathcal{O}(h^3)$  &  $\mathcal{O}(h^2)$  & 2 \\
\hline
Simpson &  $\mathcal{O}(h^5)$  &  $\mathcal{O}(h^4)$  & 4 \\
\hline
Euler &  $\mathcal{O}(h^2)$  &  $\mathcal{O}(h)$  & 1 \\
\hline
\end{tabular}
\end{table}
```

Marking notes. The second header row starts with a bare `&` and ends with `& \\\` — those two cells are *already occupied* by the `\multirow` above. Forgetting the empty cells is the single most common failure. `\cline{2-3}` rules only the two Error columns; a full `\hline` there would cut through “Method” and “Order”.

Solution 3

```

\begin{table}[H]
\centering
\caption{Standard derivatives and integrals by family.}
\renewcommand{\arraystretch}{1.7}
\begin{tabular}{|c|c|c|c|}
\hline
\multirow{2}{*}{\textbf{Family}} & & & \\
\multicolumn{2}{c|}{\textbf{Operation}} & & \\
\multicolumn{2}{c|}{\textbf{Domain}} & \& \\
\cline{2-3}
& \textbf{Function} & \textbf{Derivative} & \\
\hline
\multirow{3}{*}{Power}
&  $x^n$  &  $x^{n-1}$  &  $n \in \mathbb{R}$  \\
&  $\frac{1}{x}$  &  $-\frac{1}{x^2}$  &  $x \neq 0$  \\
&  $\sqrt{x}$  &  $\frac{1}{2\sqrt{x}}$  &  $x > 0$  \\
\hline
\multirow{2}{*}{Trigonometric}
&  $\sin x$  &  $\cos x$  &  $x \in \mathbb{R}$  \\
&  $\tan x$  &  $\sec^2 x$  &  $x \neq \frac{\pi}{2}$  \\
\hline
\multirow{2}{*}{Exponential}
&  $e^{ax}$  &  $a e^{ax}$  &  $a \in \mathbb{R}$  \\
&  $\ln x$  &  $\frac{1}{x}$  &  $x > 0$  \\
\hline
\multicolumn{2}{c|}{\textbf{Chain rule }  $f(g(x))$ } & & \\
&  $f'(g(x)) \cdot g'(x)$  & & differentiable \\
\hline
\end{tabular}
\end{table}

```

Marking notes. This is the Laplace-table pattern with different content. Three rules to internalise: (i) after a `\multirow{3}`, the next *two* rows must begin with a bare `&`; (ii) use `\cline{2-4}` between sub-rows so the family label stays unbroken, and a full `\hline` only when the family changes; (iii) without `\renewcommand{\arraystretch}{1.7}` the stacked `\dfrac`s collide. The last row flips orientation — `\multicolumn{2}{c|}` merges *across* instead of down.

Solution 4

```

\begin{table}[H]
\centering
\caption{Semiconductor material parameters.}
\renewcommand{\arraystretch}{1.5}
\begin{tabular}{|l|cc|cc|}
\hline
\rowcolor{Gray!35}
\multirow{2}{*}{\textbf{Material}} & & & & \\
\multicolumn{2}{c|}{\textbf{Electrical}} & & & \\
\multicolumn{2}{c|}{\textbf{Derived}} & & & \\
\cline{2-5}
\rowcolor{Gray!20}
&  $E_g$  (eV) &  $\mu_n$  &  $\sigma = \frac{1}{\rho}$  &  $\beta = \sqrt{\mu_n}$  \\
\hline
\rowcolor{Yellow!15} Silicon & 1.12 &  $1.35 \times 10^3$  &  $4.35 \times 10^{-4}$  & 36.7 \\
\rowcolor{Yellow!25} Germanium & 0.67 &  $3.90 \times 10^3$  &  $2.17 \times 10^{-2}$  & 62.4 \\
\rowcolor{Cyan!15} GaAs & 1.42 &  $8.50 \times 10^3$  &  $1.00 \times 10^{-8}$  & 92.2 \\
\rowcolor{Cyan!25} InP & 1.34 &  $5.40 \times 10^3$  &  $1.25 \times 10^{-7}$  & 73.5 \\
\hline
\multicolumn{3}{|r|}{\textbf{Mean gap}} & & \\
\multicolumn{2}{c|}{\textbf{Mean gap}} & & 1.14 eV & \\
\hline
\end{tabular}

```

```
\hline
\end{tabular}
\end{table}
```

Marking notes. `\rowcolor` must be the **first token of the row**, before `\multirow` — anywhere else and it errors. The colour package must be loaded as `\usepackage[table]{xcolor}`; without the `[table]` option `\rowcolor` is undefined. `Gray!35` means 35% Gray on white. Note two `\multicolumns` coexist in the final row, each carrying its own `\cellcolor`.

Solution 5

```
\begin{table}[H]
\centering
\caption{Course assessment breakdown.}
\renewcommand{\arraystretch}{1.4}
\begin{tabular}{|c|l|c|c|}
\hline
\textbf{Component} & \textbf{Item} & \textbf{Weight} & \textbf{Marks} \\
\hline
\multirow{3}{*}{Continuous}
& Quizzes & 10\% & 10 \\
& Assignments & 15\% & 15 \\
& Lab work & 15\% & 15 \\
\hline
\multirow{2}{*}{Examination}
& Midterm & 25\% & 25 \\
& Final & 35\% & 35 \\
\hline
\multicolumn{2}{|c|}{\textbf{Total}} & \textbf{100\%} & \textbf{100} \\
\hline
\end{tabular}
\end{table}
```

Marking notes. The percent sign must be escaped as `\%` — a bare `%` comments out the rest of the line and silently eats your row. This is a favourite way to lose marks. The footer merges the first two columns while the last two stay separate.

Solution 6

```
\begin{table}[H]
\centering
\caption{Comparison of transform methods.}
\renewcommand{\arraystretch}{1.2}
\begin{tabular}{lcc}
\toprule
\textbf{Transform} & \textbf{Kernel} & \textbf{function} \\
\midrule
\textbf{Typical} & \textbf{application} & \\
\midrule
Laplace &  $e^{-st}$  & transient circuits, control systems \\
Fourier &  $e^{-j\omega t}$  & spectra, signal filtering \\
Z &  $z^{-n}$  & discrete systems, digital filters \\
\midrule
\multicolumn{3}{c}{\textit{All three convert differential relations into algebraic ones.}} \\
\bottomrule
\end{tabular}
\end{table}
```

Marking notes. booktabs style deliberately has *no* vertical rules, so the column spec is plain {lcc}. Never mix \hline with \toprule/\midrule. \makecell{a\\b} is how you break a line inside a cell — a bare \\ there would end the whole row instead.

Solution 7

```
\begin{table}[H]
\centering
\caption{Boundary conditions and their meaning.}
\renewcommand{\arraystretch}{1.3}
\begin{tabularx}{0.92\textwidth}{|l|c|X|}
\hline
\textbf{Name} & \textbf{Form} & \textbf{Physical interpretation} \\
\hline
Dirichlet &  $u = g$  on  $\partial\Omega$  & The value of the solution is prescribed on the boundary, for example a surface held at a fixed temperature. \\
\hline
Neumann &  $\frac{\partial u}{\partial n} = g$  & The flux across the boundary is prescribed; a zero value models a perfectly insulated wall. \\
\hline
Robin &  $\alpha u + \beta \frac{\partial u}{\partial n} = g$  & A weighted combination of the two above, used for convective heat exchange with a surrounding medium. \\
\hline
\end{tabularx}
\end{table}
```

Marking notes. tabularx takes an *extra* first argument: the total width. The X column absorbs the leftover space and wraps text automatically — this is the fix whenever a table overflows the margin. You may use several X columns; they share the space equally.

Solution 8

```
\begin{table}[H]
\centering
\caption{Objects that must survive inside a cell.}
\renewcommand{\arraystretch}{2.0}
\begin{tabular}{|c|c|c|}
\hline
\textbf{Object} & \textbf{Expression} & \textbf{Note} \\
\hline
Matrix &  $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$  &  $2 \times 2$  real \\
\hline
Determinant &  $\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$  & identity \\
\hline
Piecewise &  $f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}$  & continuous at 0 \\
\hline
Integral &  $\int_0^\infty e^{-st} dt = \frac{1}{s}$  &  $s > 0$  \\
\hline
Limit &  $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$  & classical \\
\hline
Nested index &  $\sum_{\substack{i=1 \\ i \neq j}}^n a_{i,j}$  & double subscript \\
\hline
\end{tabular}
```

```

\hline
\end{tabular}
\end{table}

```

Marking notes. The surprise here is that `&` and `\\` inside `bmatrix/cases` do *not* break the outer table — the inner environment claims them. That is why the matrix row works despite containing three `&` and one `\\`. Add `\displaystyle` to force full-size $\sum$, $\int$ and $\lim$ with limits above and below; without it they shrink to inline size. Push `\arraystretch` up to 2.0 to stop tall cells colliding.

Solution 9

```

\begin{table}[H]
\centering
\caption{Network layer performance matrix.}
\renewcommand{\arraystretch}{1.4}
\begin{tabular}{|c|c|c|c|c|}
\hline
\multirow{3}{*}{\diagbox[width=6em]{\textbf{Layer}}{\textbf{Metric}}} &
\multicolumn{2}{c|}{\textbf{Training}} & &
\multicolumn{2}{c|}{\textbf{Inference}} \\
\cline{2-5}
& \multicolumn{2}{c|}{(per epoch)} & & \multicolumn{2}{c|}{(per sample)} \\
\cline{2-5}
& \textbf{Time} & \textbf{Memory} & \textbf{Time} & \textbf{Memory} \\
\hline
Convolution & $12.4$ s & $1.8$ GB & $3.1$ ms & $0.4$ GB \\
\hline
Pooling & $0.8$ s & $0.2$ GB & $0.2$ ms & $0.1$ GB \\
\hline
Dense & $4.6$ s & $3.4$ GB & $1.0$ ms & $0.9$ GB \\
\hline
\multicolumn{1}{|r|}{\textbf{Total}} & $17.8$ s & $5.4$ GB & $4.3$ ms & $1.4$ GB \\
\hline
\end{tabular}
\end{table}

```

Marking notes. A `\multirow{3}` means the next *two* rows each open with a bare `&`. `\diagbox{A}{B}` (package `diagbox`) splits one cell with a diagonal; fix its width with the optional `[width=6em]` or the column will bulge. If `diagbox` is unavailable on the exam machine, substitute `\multirow{3}{*}{\textbf{La}}` and move on — structure earns more marks than the diagonal. The last row uses `\multicolumn{1}{r|}` purely to *re-align* a single cell to the right, which is a legitimate and handy trick.

Solution 10

Table~\ref{tab:sim} summarises the four candidate designs.

```

\begin{table}[H]
\centering
\caption{Design trade-off study for the RC filter stage.}
\label{tab:sim}
\renewcommand{\arraystretch}{1.6}
\begin{tabular}{|l|c|c|c|c|}
\hline
\rowcolor{RoyalBlue!25}
\multirow{2}{*}{\textbf{Design}} & & & & \\
\multicolumn{2}{c|}{\textbf{Components}} & & & \\
\multicolumn{2}{c|}{\textbf{Response}} & & & \\
\hline
\end{tabular}

```

```

\cline{2-5}
\rowcolor{RoyalBlue!12}
& $\mathit{R}$ (k$\Omega$) & $\mathit{C}$ ($\mu\mathit{F}$) &
  $\tau$ = RC & $f_c$ = $\frac{1}{2\pi RC}$ \\
\hline
\multirow{2}{*}{Fast}
& 1.0 & 0.10 & $0.10$ ms & $1.59$ kHz \\ \cline{2-5}
& 2.2 & 0.10 & $0.22$ ms & $723$ Hz \\
\hline
\multirow{2}{*}{Slow}
& 10.0 & 1.00 & $10.0$ ms & $15.9$ Hz \\ \cline{2-5}
& 22.0 & 4.70 & $103$ ms & $1.54$ Hz\footnotemark \\
\hline
\rowcolor{Green!12}
\multicolumn{3}{|r|}{\textbf{Selected design}} & &
  \multicolumn{2}{c|}{Fast, $\mathit{R}$ = $1.0$ k$\Omega$} \\
\hline
\end{tabular}
\end{table}
\footnotetext{Measured with a $5$V step input at $25^\circ\mathit{C}$.}

The selected design in Table~\ref{tab:sim} gives the widest bandwidth.

```

Marking notes. Five things are graded at once here. `\caption` goes *before* `\label`, and `\label` must follow `\caption` or `\ref` prints the wrong number. `\ref{tab:sim}` resolves only on the **second** compile — a ?? means run again, not that your code is wrong. A normal `\footnote` is swallowed inside a float, so use `\footnotemark` in the cell and `\footnotetext` just after the table. Greek and units inside cells are ordinary maths: `k$\Omega$`, `$\mu\mathit{F}$`, `$25^\circ\mathit{C}$`. And `\rowcolor` still has to lead its row even when a `\multicolumn` follows.

Drill Routine

1. **Round 1 — untimed.** Work through 1–5 with this file’s Part II open. The aim is recognition, not speed.
2. **Round 2 — timed, easy half.** Close Part II. Reproduce 1, 2, 5, 6 in under 20 minutes total. If you exceed it, the bottleneck is almost always counting columns — write the column spec first and count the |.
3. **Round 3 — timed, hard half.** Reproduce 3, 4, 8 in under 25 minutes. These three contain every pattern the real papers used.
4. **Round 4 — exam pace.** Do Exercise 10 cold in 15 minutes, including the surrounding sentence and the cross-reference.
5. **Round 5 — from memory.** Write Exercise 3 with no reference open at all. When you can do that, tables will not cost you marks.

The Five Errors That Cost the Most Marks

#	Mistake	Symptom and fix
1	Missing <code>&</code> after a <code>\multirow</code>	Extra alignment tab or shifted columns. Every row covered by a <code>\multirow {N}</code> needs a placeholder <code>&</code> in that column.
2	<code>\rowcolor</code> not first	Error or uncoloured row. It must precede <code>\multirow</code> and <code>\multicolumn</code> .
3	Bare A row silently vanishes. Escape them: <code>\verb% </code> and <code>\&</code> .	
4	Column count mismatch	Extra alignment tab has been changed to <code>\cr</code> . Count the <code>&</code> in the row: it must be one fewer than the number of columns, after accounting for spans.
5	Compiling once	<code>\ref</code> shows ?? and the TOC is stale. Always run <code>pdflatex</code> twice.

Reproduce, compile, compare, repeat.